# Robotics



Mahmoud Abdelaal

# Lecture # 2

# Robotics Fundamentals

# Industrial Robotic Arm

# (Manipulator)

# Industrial Robotic Arm

**Manipulators / Robotic Arms / Industrial Robots**





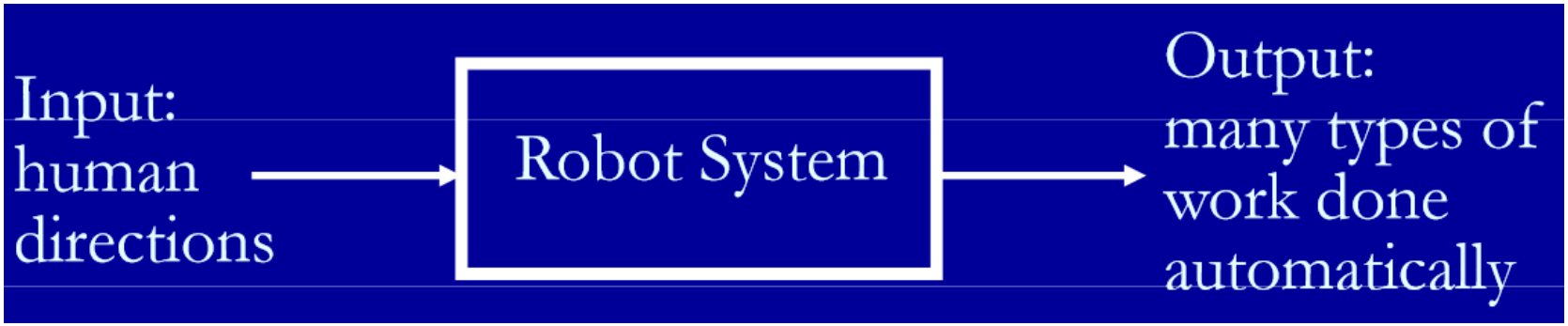






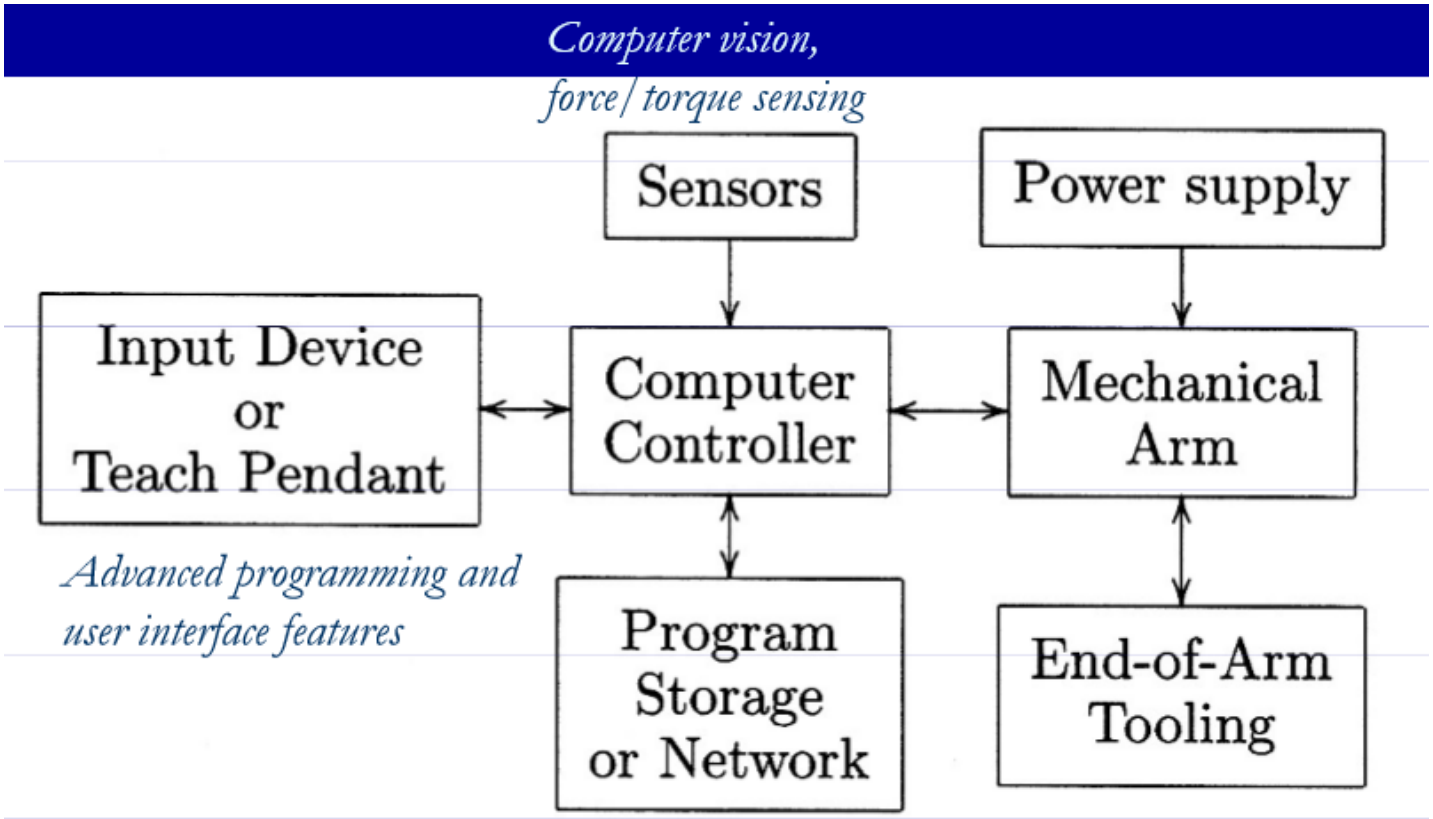

# Industrial Robotic Arm

**Robot System as a Black Box**



# Industrial Robotic Arm

## A Robotic System



# Industrial Robotic Arm

**Networked Robots**

- A robotic device connected to a communications network such as the Internet or LAN.
- The network could be wired or wireless, and based on any of a variety of protocols such as TCP, UDP, ….

# Industrial Robotic Arm

**Accuracy vs. Repeatability vs. Resolution**

**Accuracy:** How close to a given point

How close the manipulator can come to a given point within its workspace (computational errors, machining accuracy, flexibility effects, gear backlash, static/dynamic effects)

**Repeatability:** How close to previously taught point.

How close a manipulator can return to a previously taught point (controller resolution)

**Resolution:**

computed as the total distance traveled divided by 2^$n$,where $n$ the number of bits of encoder accuracy. (the smallest increment of motion that the controller can sense)

# Robot Components

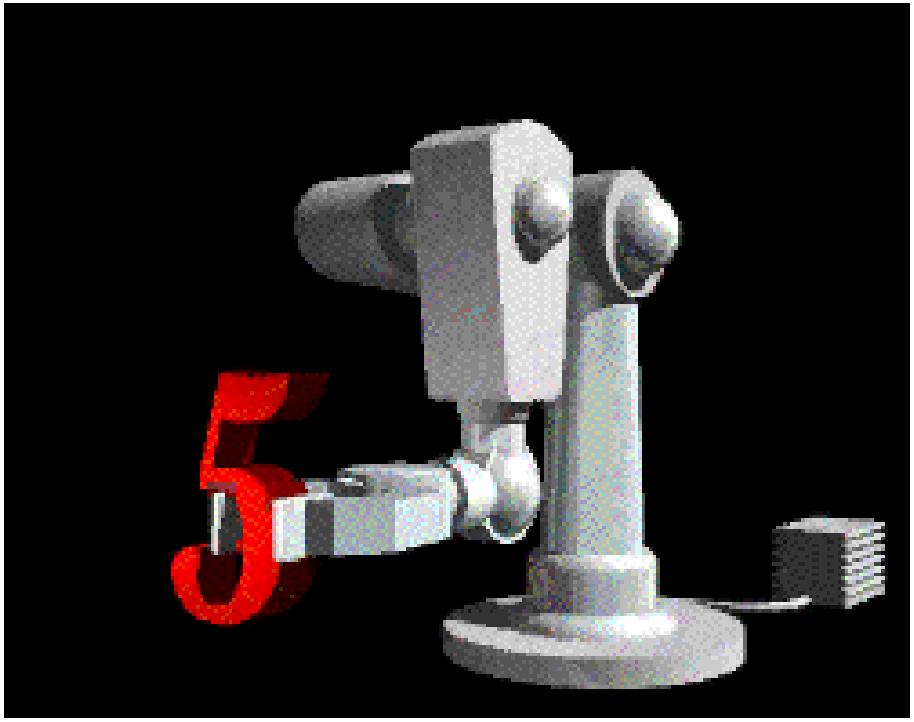
# Robot Components

- **Manipulator / Arm**
- **End Effector:** is connected to the last joint (ex: Hand / Gripper)
  - Hold the part while work is being performed
  - Hold tool which is performing work on the part
- **Actuators :** Actuators are the muscles of the manipulators. As servomotors, Stepper motors, pneumatic and hydraulic .
- **Sensors:** are used to collect information to communicate with the environment.
- **Controller:** receive its data from the computer and controls the motion of the actuators.
- **Processor:** calculate the motion of the joints.
- **Software :** there are groups of software programs used in robot

# Robot Components

**What are the parts of a robot?**

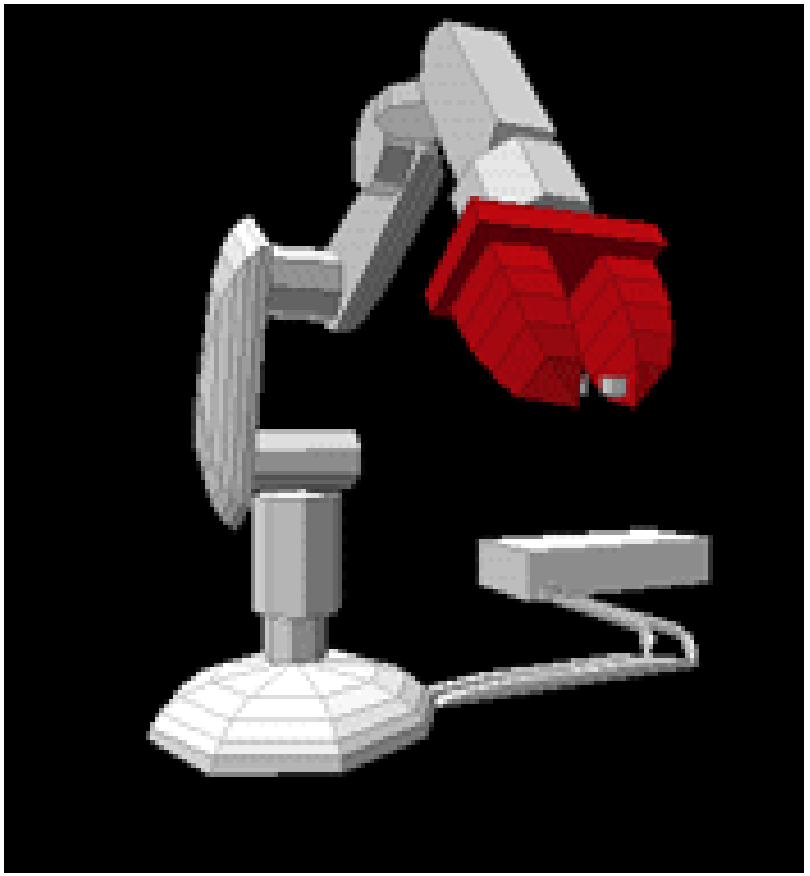
- Manipulator
- Pedestal
- Controller
- End Effectors
- Power Source



# Robot Components

**Manipulator**

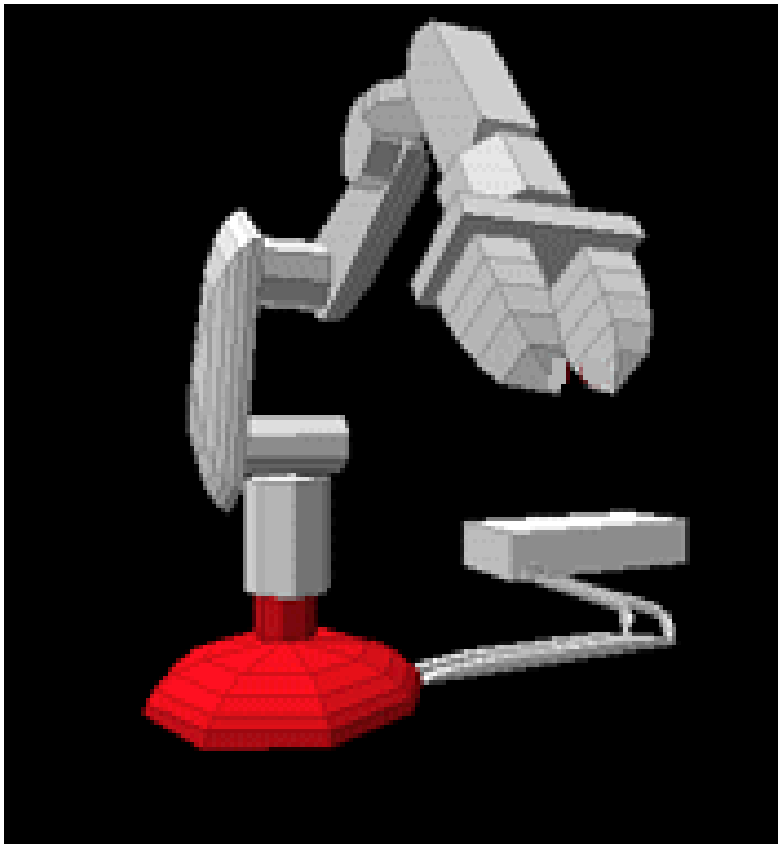
- Base
- Appendages
  - Shoulder
  - Arm
  - Grippers



# Robot Components

**Pedestal (Human waist)**

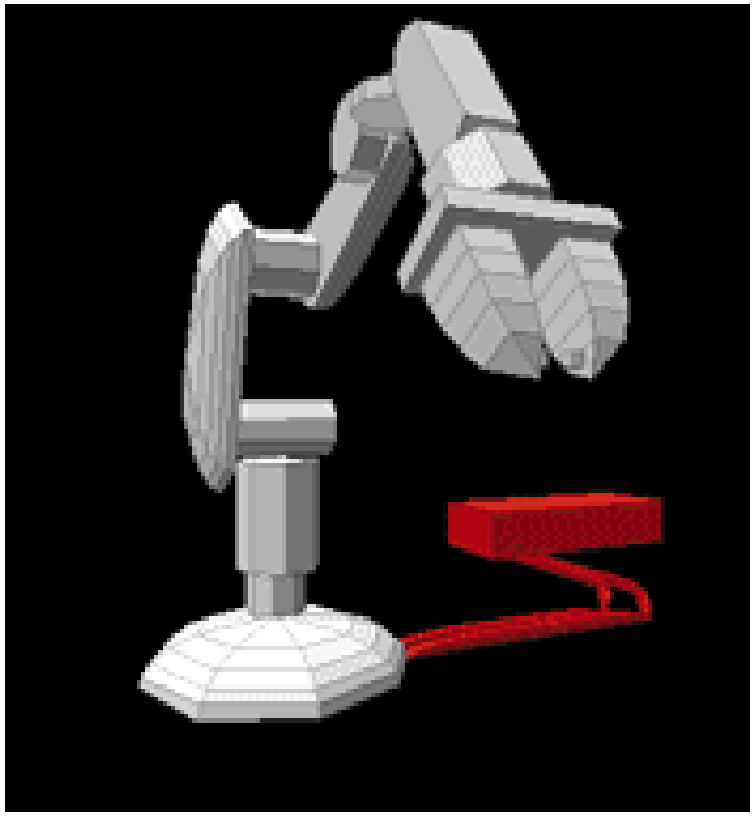
- Supports the manipulator.
- Acts as a counterbalance.



# Robot Components

**Controller (The brain)**

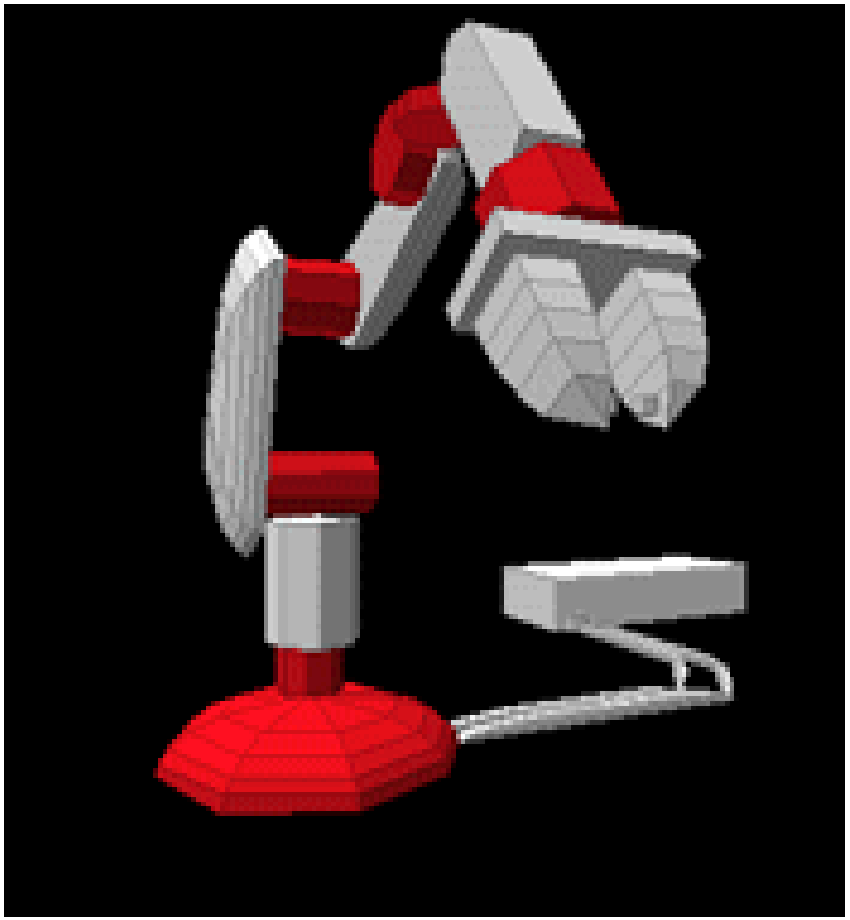
- Issues instructions to the robot.
- Controls peripheral devices.
- Interfaces with robot.
- Interfaces with humans.



# Robot Components

**Power Source (The food)**

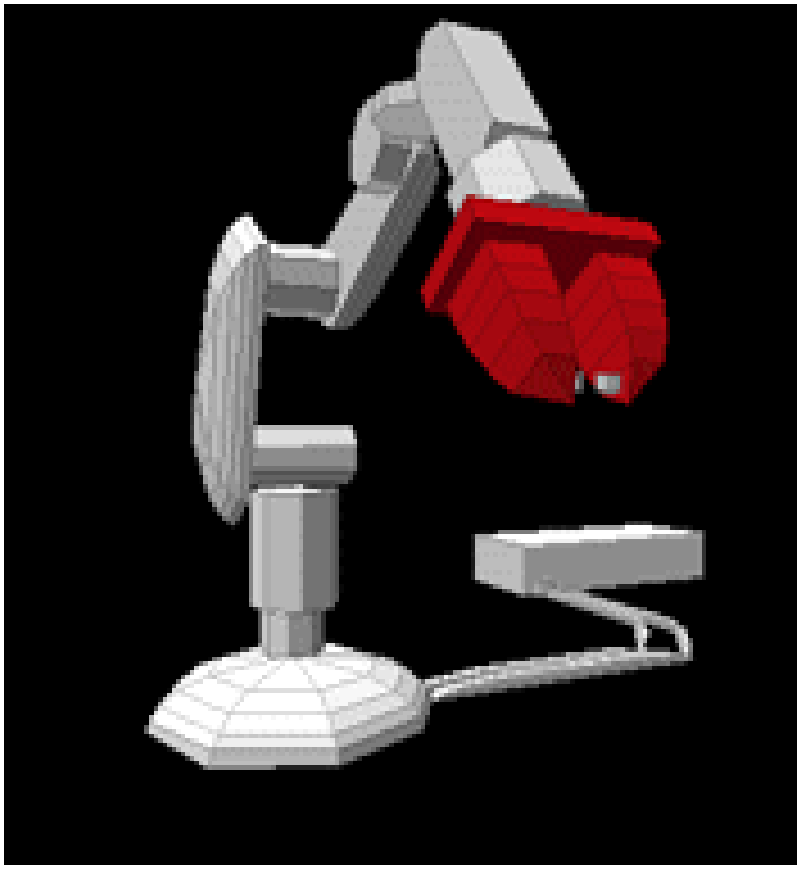
- Electric
- Pneumatic
- Hydraulic



# Robot Components

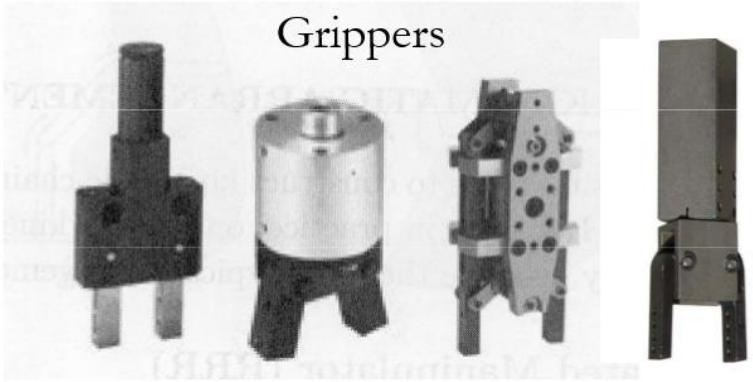
**End Effectors (The hand)**

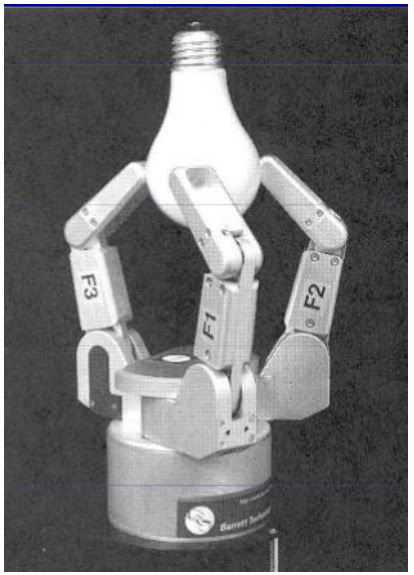
- Spray paint attachments
- Welding attachments
- Vacuum heads
- Hands
- Grippers

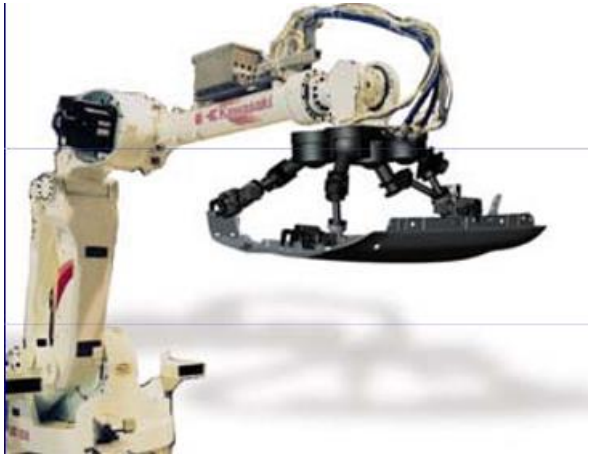


# Robot Components

## End-effectors



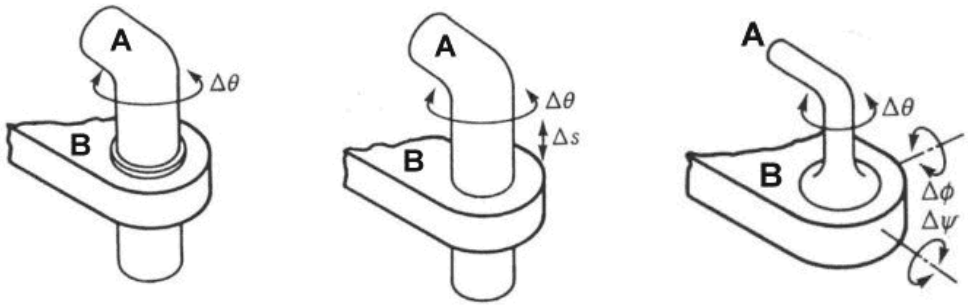



# Degrees of Freedom, DOF

# Degrees of Freedom, DOF

**Robots degrees of freedom**

- Degrees of Freedom: Number of independent position variables which would has to be specified to locate all parts of a mechanism.
- In most manipulators this is usually the number of joints.



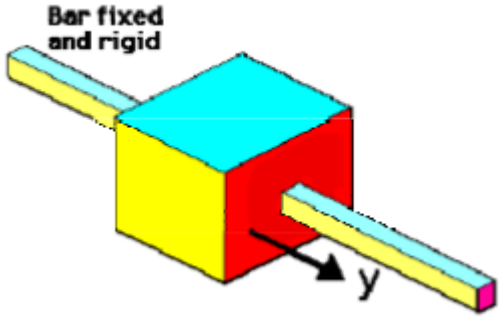

1 D.O.F. 2 D.O.F. 3 D.O.F.

# Degrees of Freedom, DOF

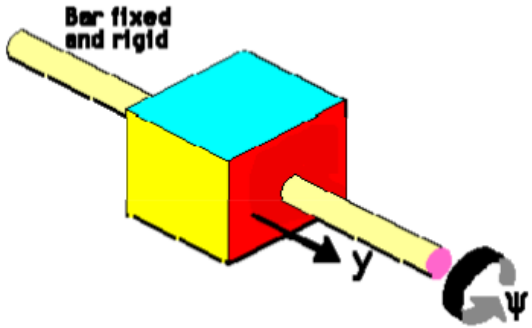
**Degree Of Freedom (DOF)**

The number of independent motions a device can make

Examples: Simple sliding



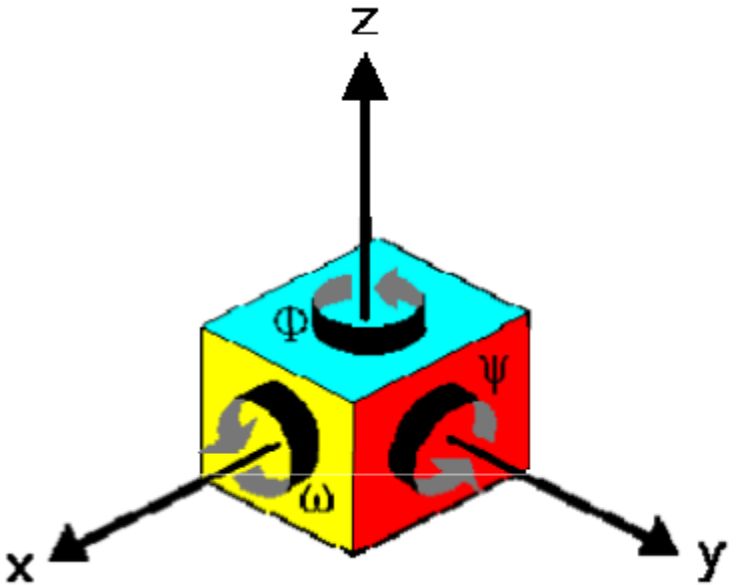

Examples: Rotation + translation:



# Degrees of Freedom, DOF

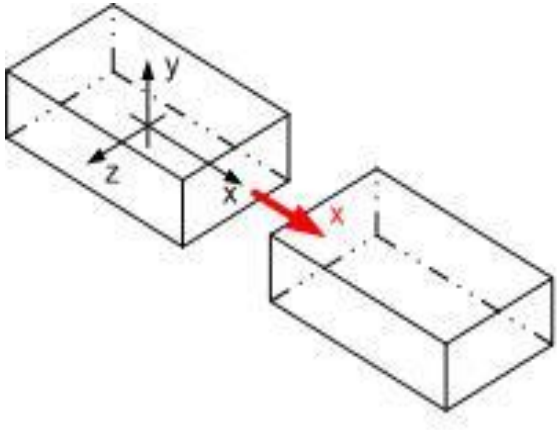
**Degree Of Freedom (DOF)**

Examples: A free body in space: 6DOF

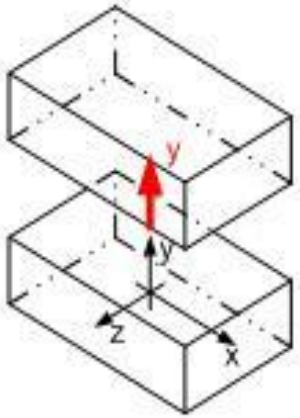


# Degrees of Freedom, DOF

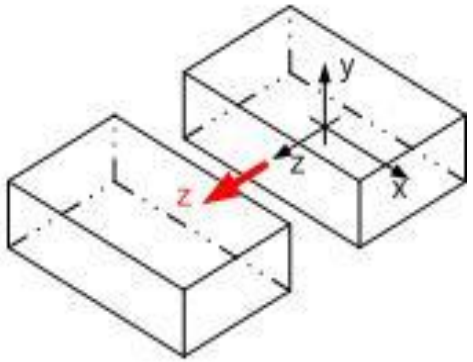
Degrees of Freedom refer to the movement range available for a given piece of equipment within three dimensions.



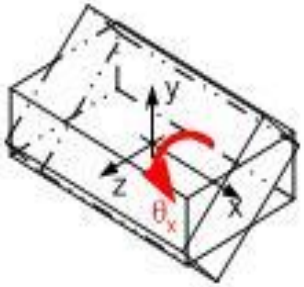

Linear in x-direction



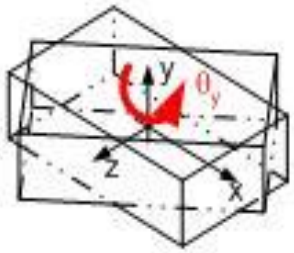

Linear in y-direction



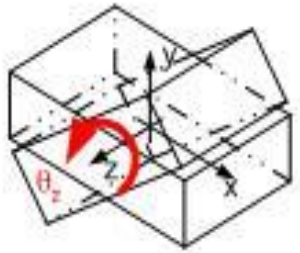

Linear in z-direction




Rotation around x-axis



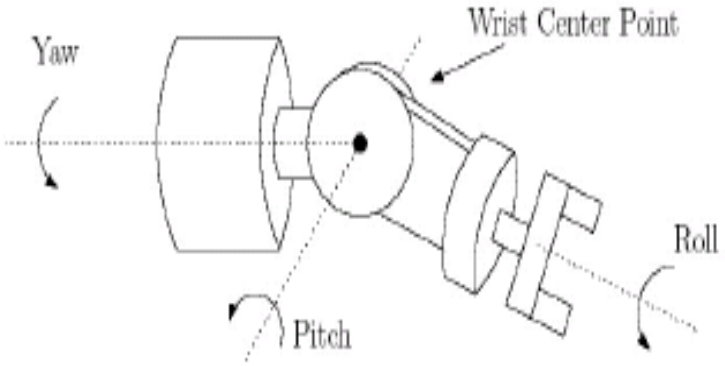

Rotation around y-axis

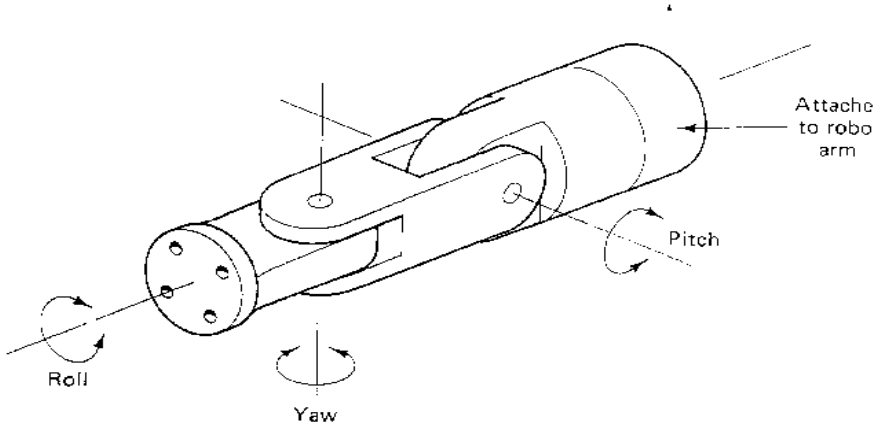



Rotation around z-axis

# Degrees of Freedom, DOF

- ✓ Roll: involves rotating the wrist about the arm axis
- ✓ Pitch: up-down rotation of the wrist
- ✓ Yaw: left-right rotation of the wrist



# Joints

# Joints

**Want to build a robot?**

- **Links:**

rigid bodies

- **Actuators:**

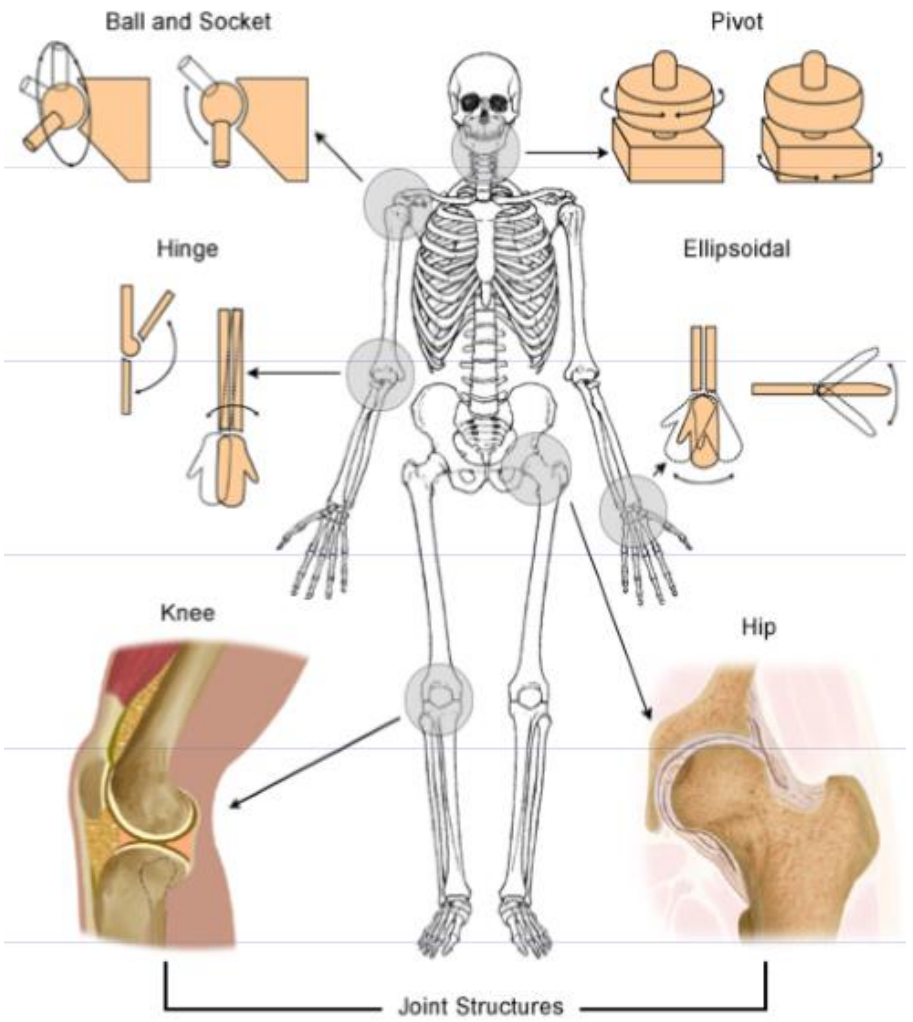
connect and move links

- **Sensors:**

perceiving the world

- **Others:**

end-effectors, communication, etc.

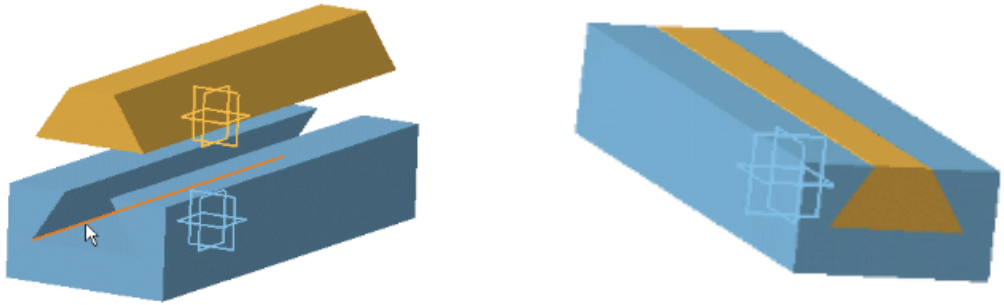


# Joints

**Robot Joints**

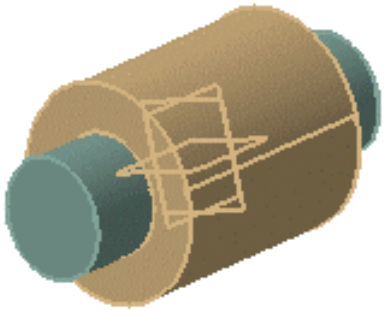
**Prismatic Joint:**

Linear, No rotation involved. (Hydraulic or pneumatic cylinder)



**Revolute Joint:**

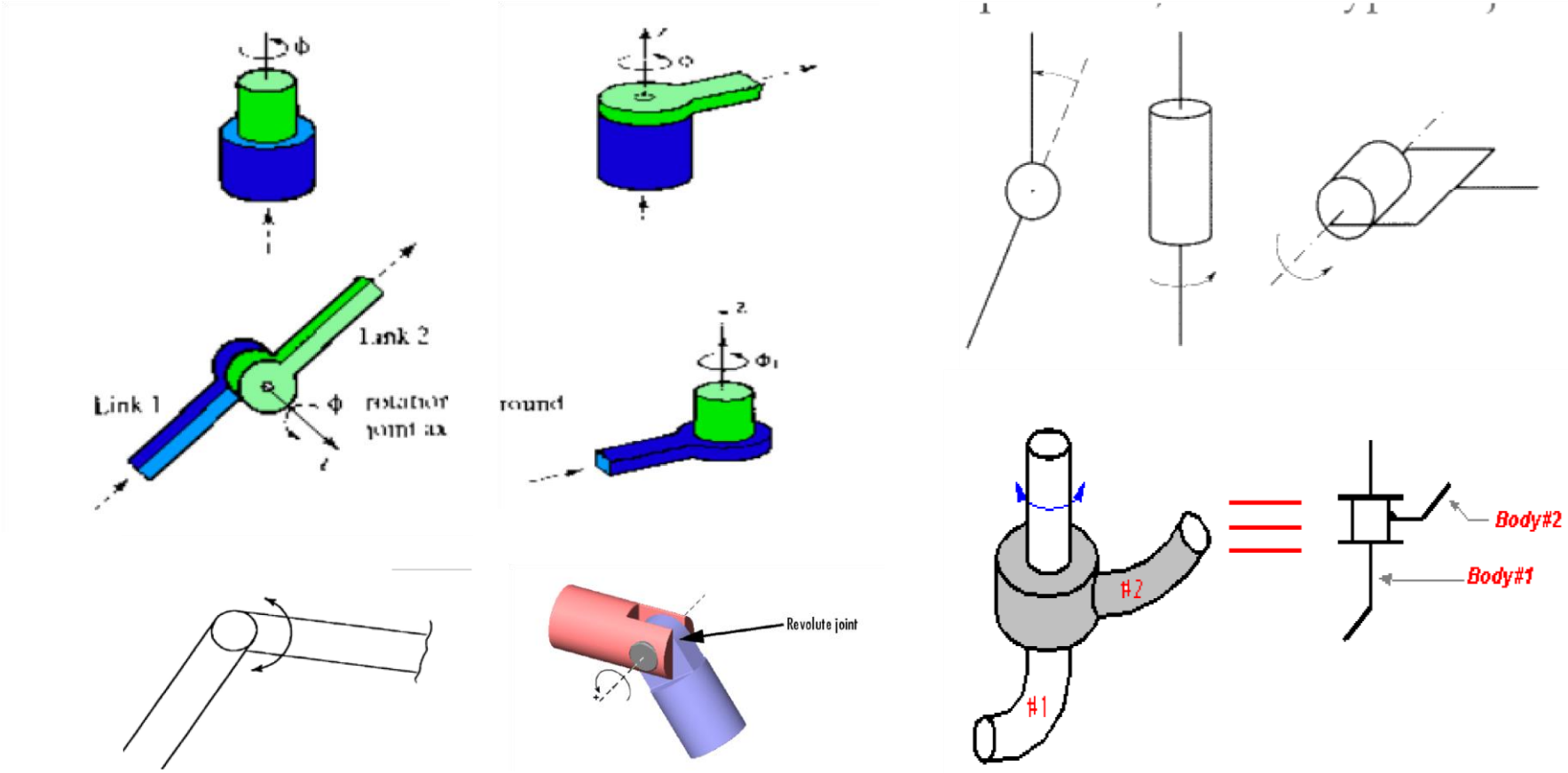
Rotary, (electrically driven with stepper motor, servo motor)



# Joints

## Revolute joint (R)

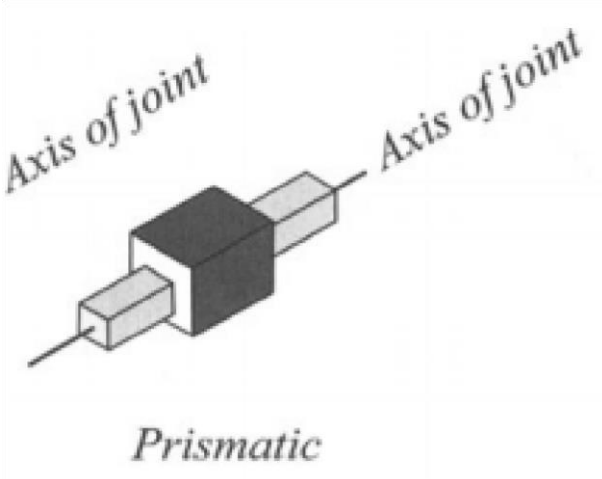
The revolute joint allows only relative rotational motion about a single axis

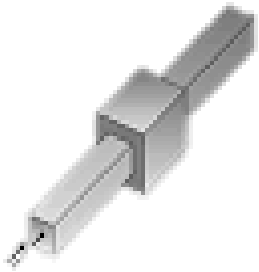


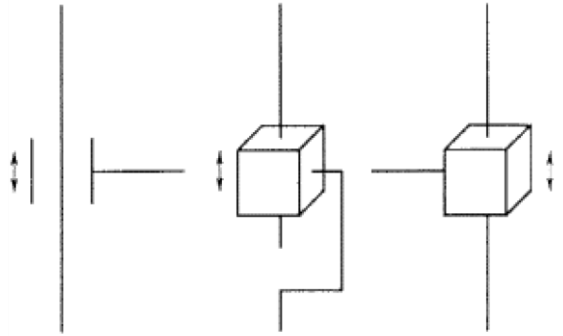
# Joints

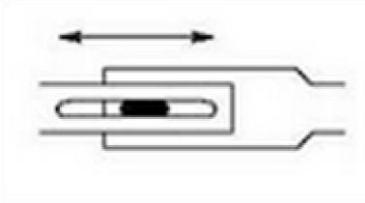
## Prismatic joint (P)

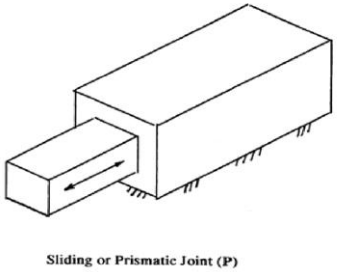
The prismatic joint allows only relative translational motion along a single axis.

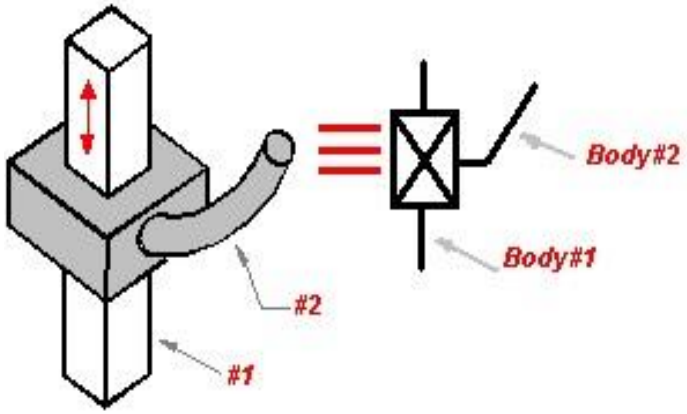




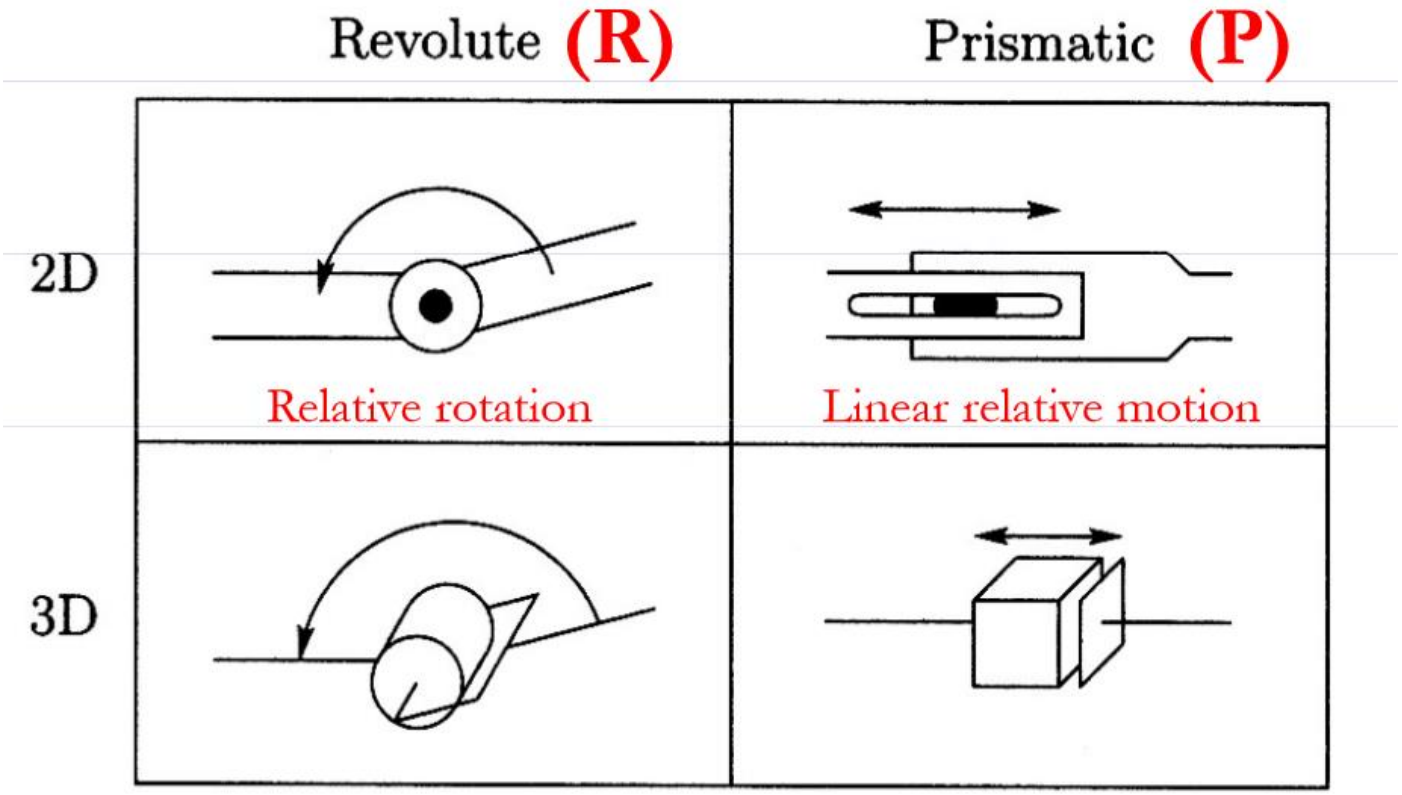




# Joints

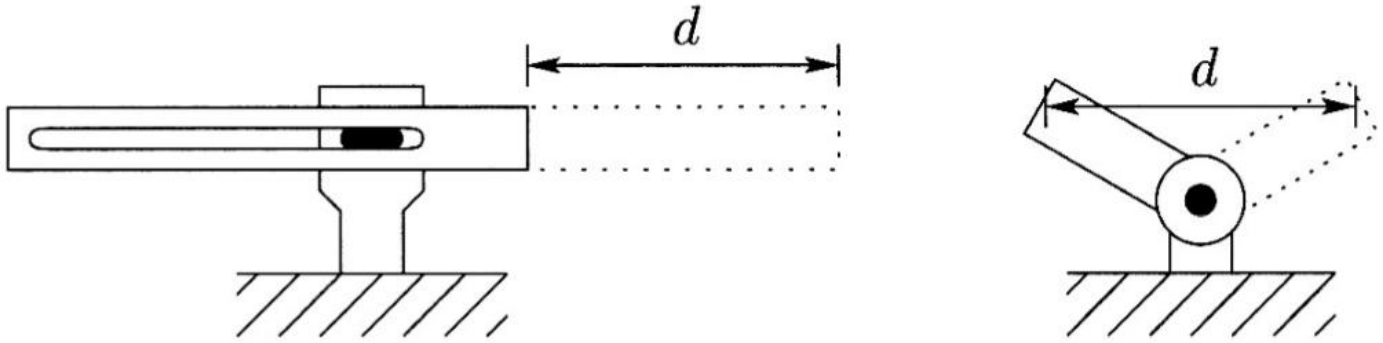
## Representation of Robot Joints



# Joints

**Linear vs. Rotational Link Motion**

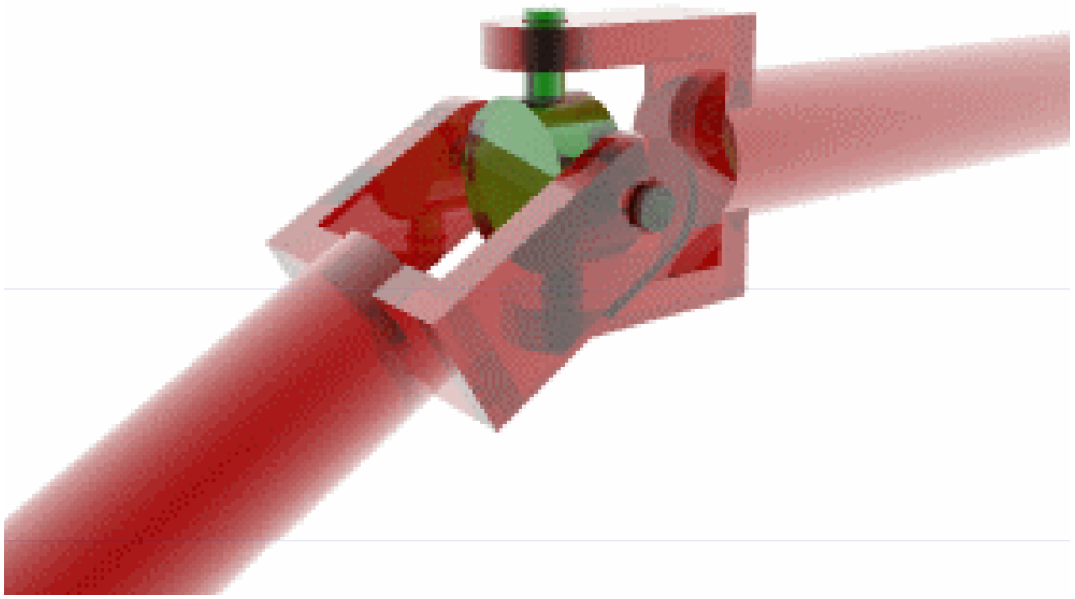
- Linear axes (prismatic joints) typically have higher resolution than revolute joint.
- Rotational axes usually result in a large amount of kinematic and dynamic coupling among the links, with a resultant accumulation of errors and a more difficult control problem.



# Joints

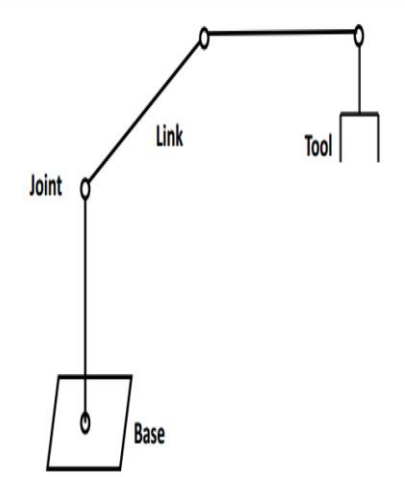
## Universal Joint

- A joint in a rigid rod that allows the rod to bend in any direction



# Remember

**Note That**



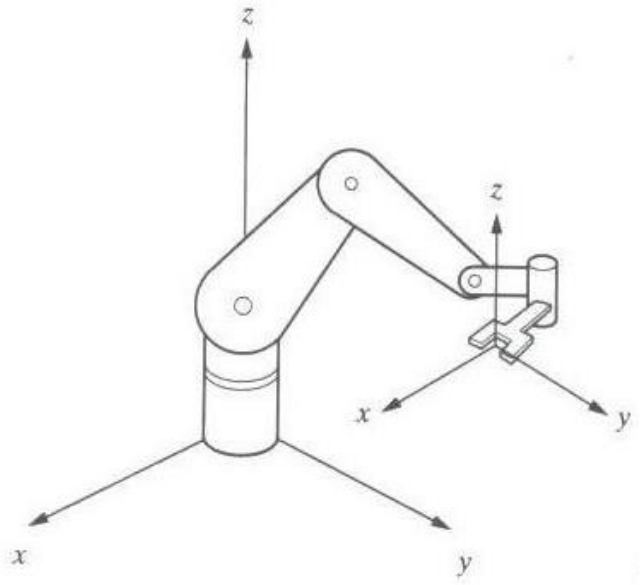

- The joint is the point of articulation between two links.
- Joint i connects links i and i-1.
- There is no joint at the end effector.
- Each joint-link pair constitutes one DOF.
- A robotic manipulator can be modeled as a chain of rigid bodies called links.
- The links are interconnected to one another by joints.
- One end of the chain of links is fixed to a base, while the other end is free to move.

# Reference Frames

# Reference Frames

**World Reference Frame**

World Reference Frame which is a universal coordinate frame, as defined by the x-y-z axes. In this case the joints of the robot move simultaneously so as to create motions along the three major axes.

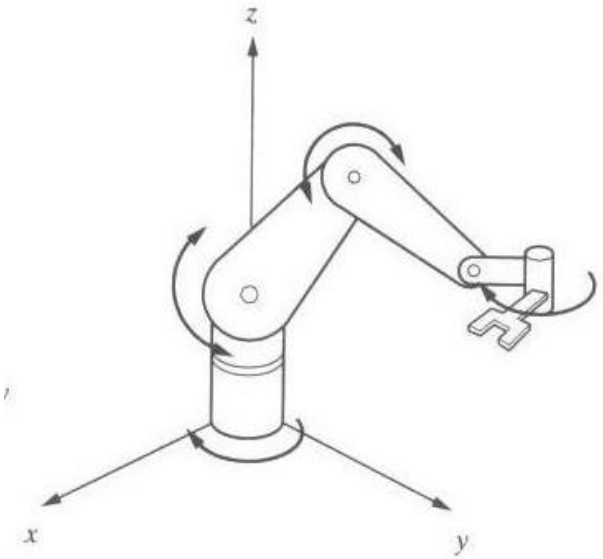



World reference frame

# Reference Frames

**Joint Reference Frame**

Joint Reference Frame which is used to specify movements of each individual joint of the Robot. In this case each joint may be accessed individually and thus only one joint moves at a time.

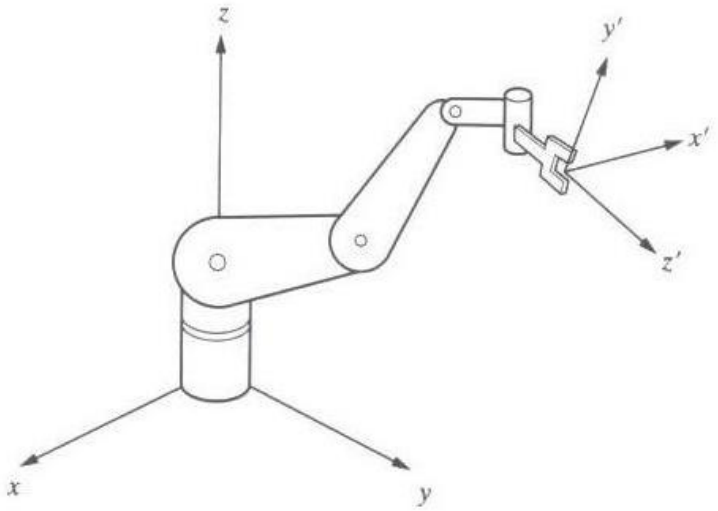



Joint reference frame

# Reference Frames

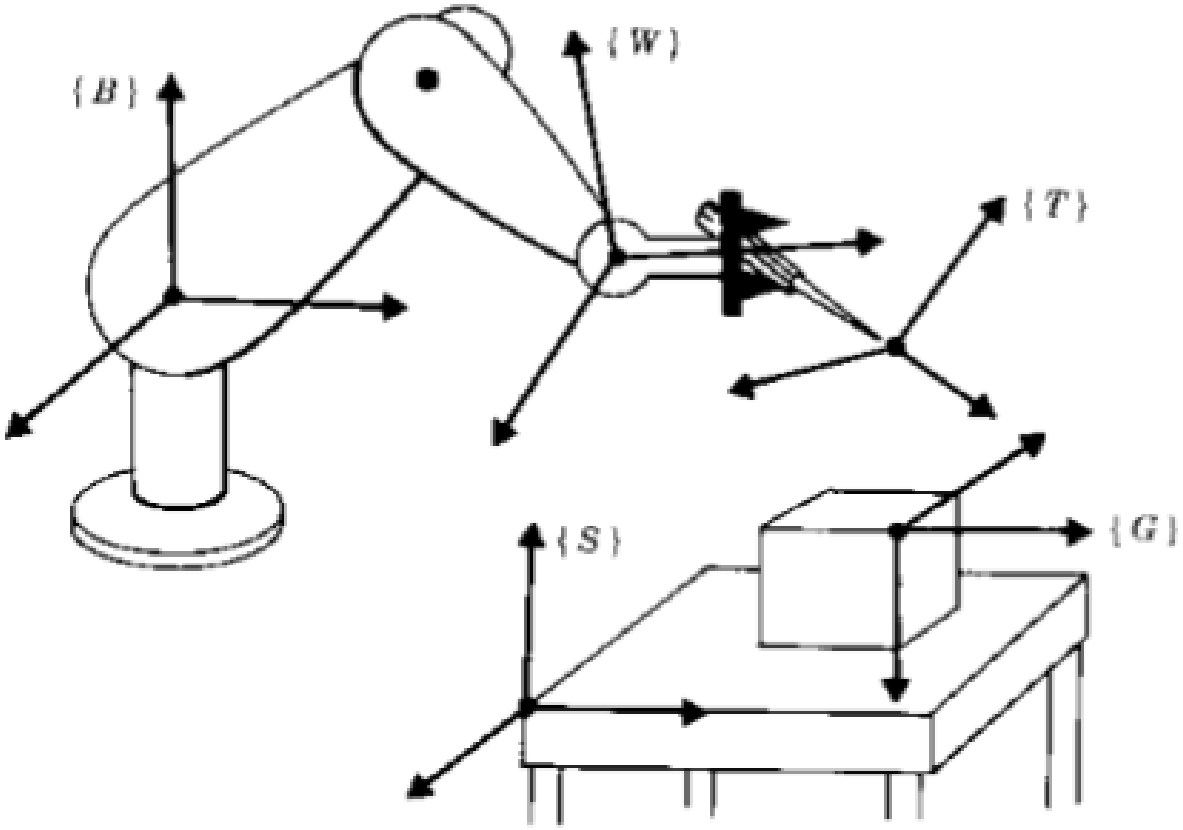
**Tool Reference Frame**

Tool Reference Frame which specifies the movements of the Robots hand relative to the frame attached to the hand. The x',y'and z' axes attached to the hand define the motions of the hand relative to this local frame. All joints of the Robot move simultaneously to create coordinated motions about the Tool frame.




Tool reference frame

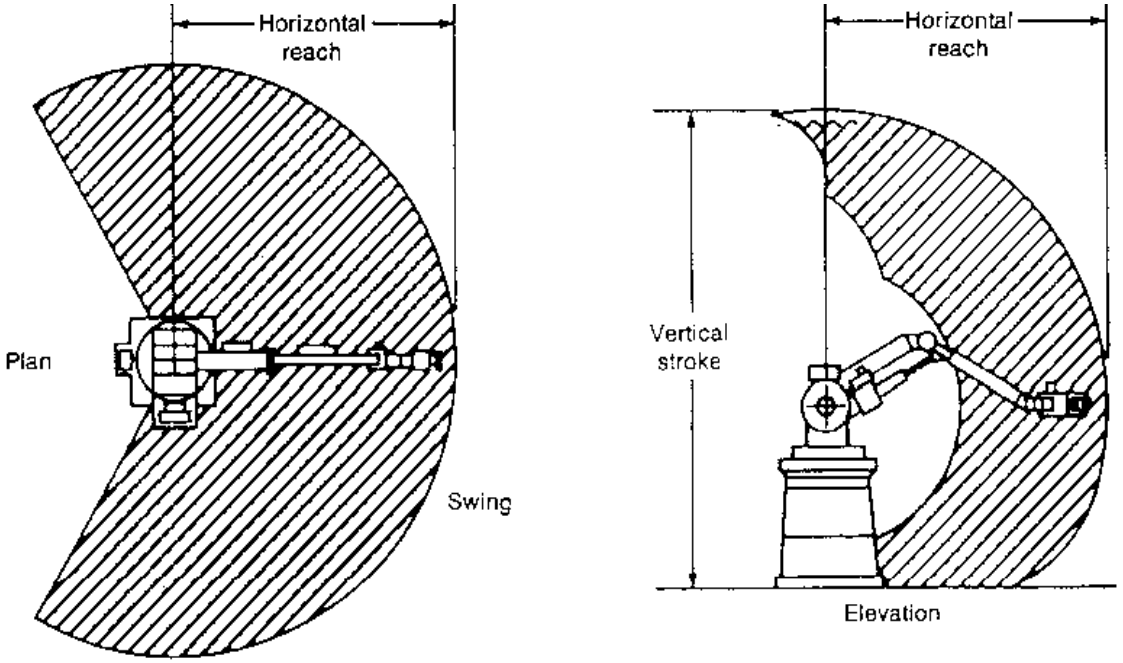
# Reference Frames



# Robot Configurations

# Work Space/Envelope concept

- Depending on the configuration and size of the links and wrist joints, robots can reach a collection of points called a Workspace.
- Alternately Workspace may be found empirically, by moving each joint through its range of motions and combining all space it can reach and subtracting what space it cannot reach

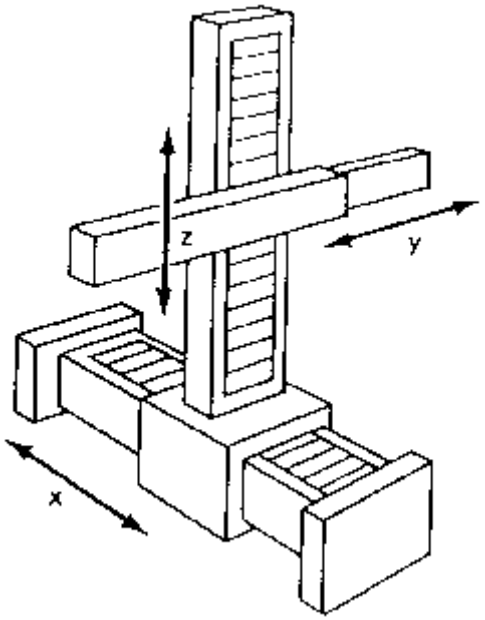


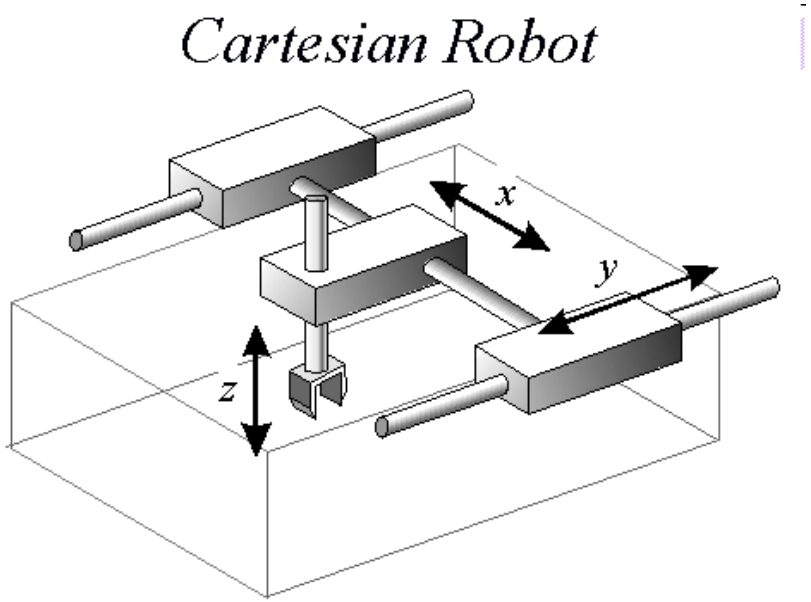
# Robot Configurations

Some of the commonly used configurations in Robotics are

**Cartesian/Rectangular Gantry(3P/ PPP) :**

These Robots are made of 3 Linear joints that orient the end effector, which are usually followed by additional revolute joints.



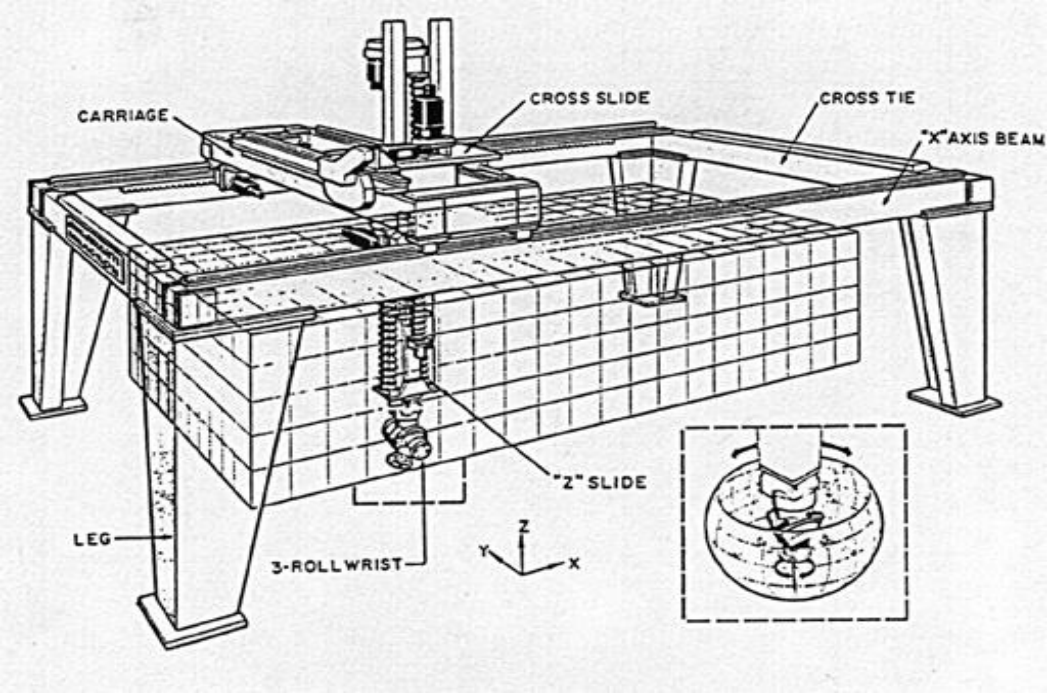



# Robot Configurations

**Cartesian Robot - Work Space/Envelope**

**Gantry**

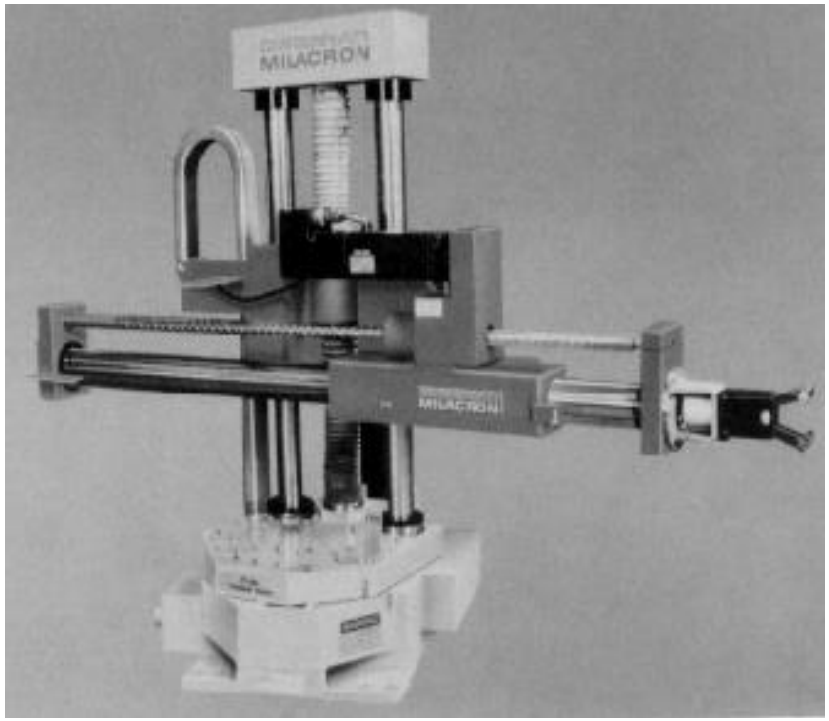
Gantry robots are cartesian robots that have been super-sized! This structure minimizes deflection along each axis.

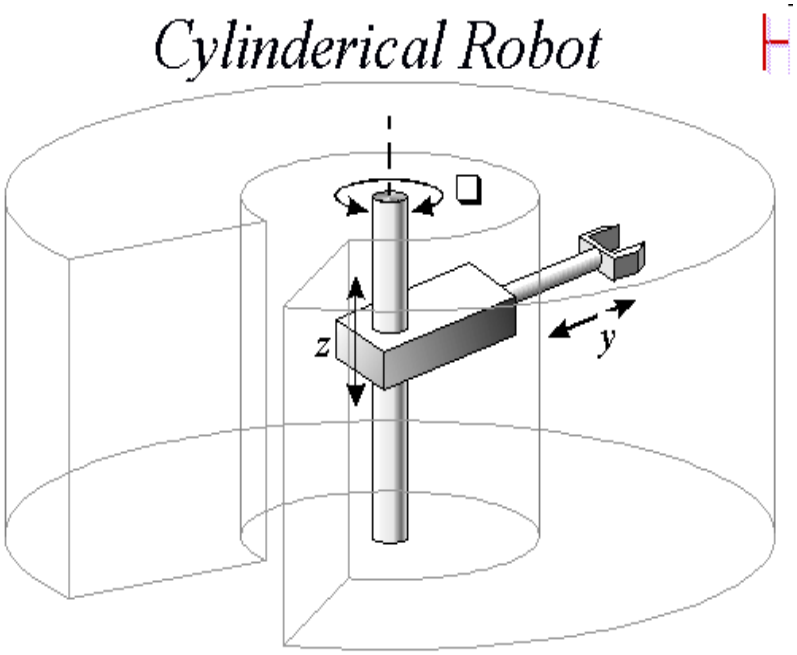


# Robot Configurations

**Cylindrical (RPP):**

Cylindrical coordinate Robots have 2 prismatic joints and one revolute joint.

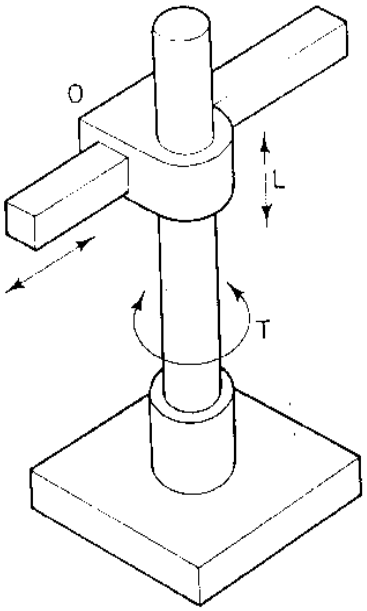


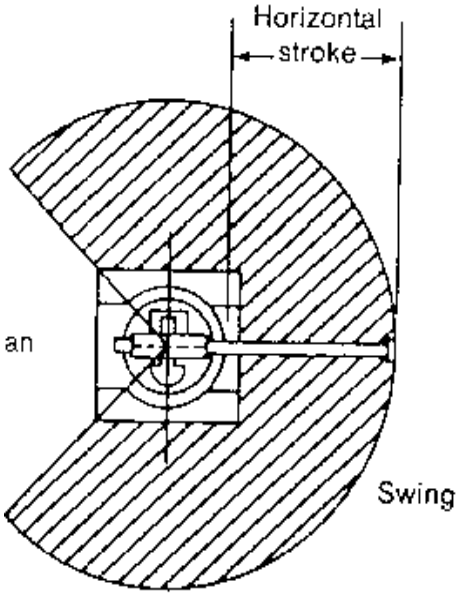



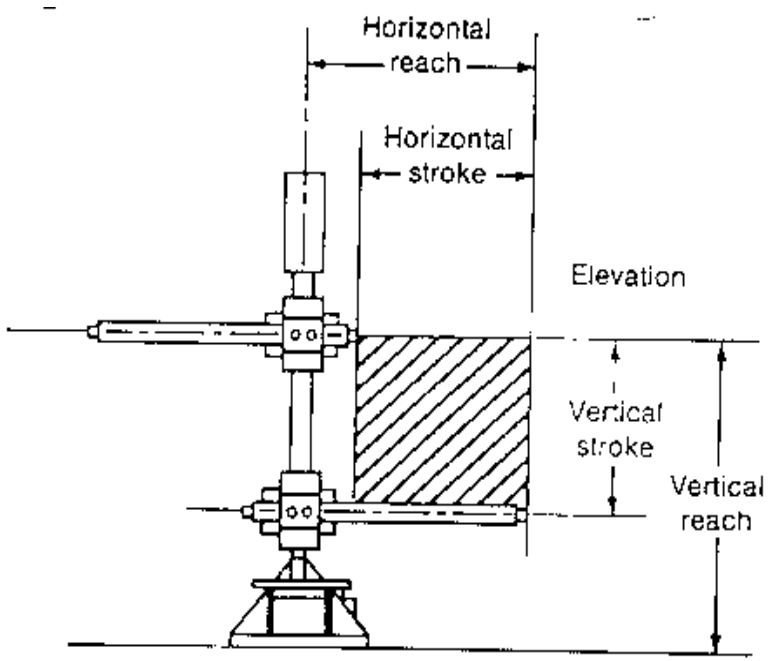
# Robot Configurations

## Cylindrical Robot - Work Space/Envelope

Cylindrical robots in the center of the have a main axis that is operating envelope. It can reach into tight areas without sacrificing speed or repeatability.



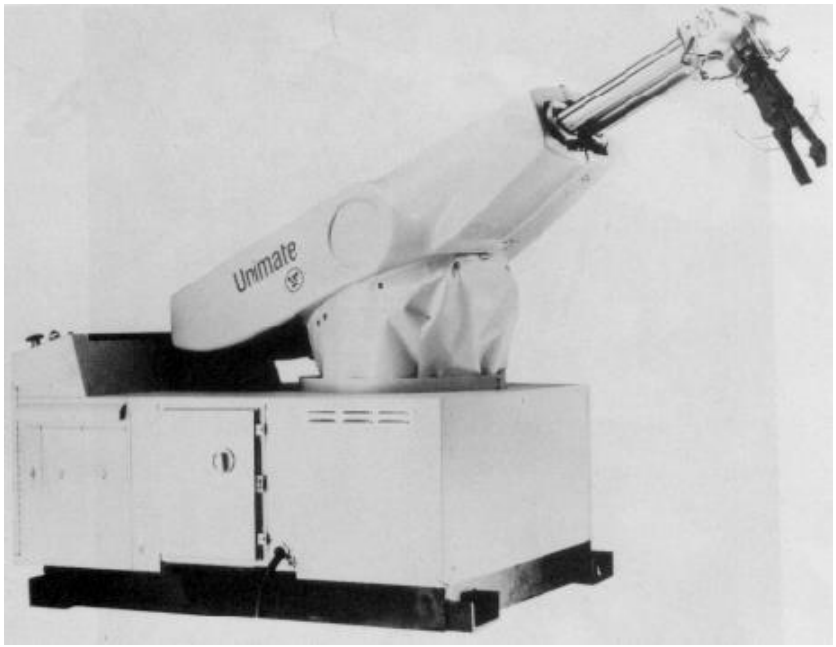



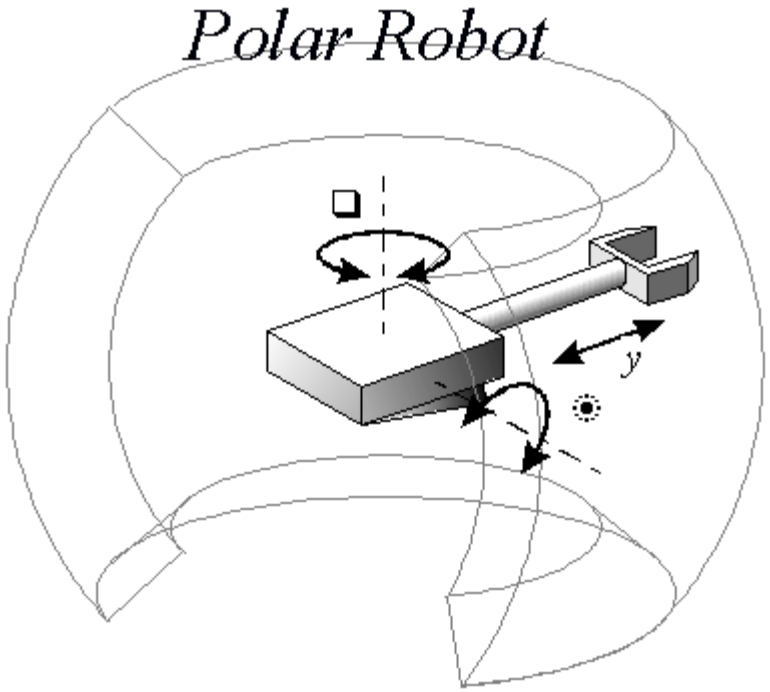



# Robot Configurations

**Spherical joint (2RP/RRP):**

They follow a spherical coordinate system, which has one

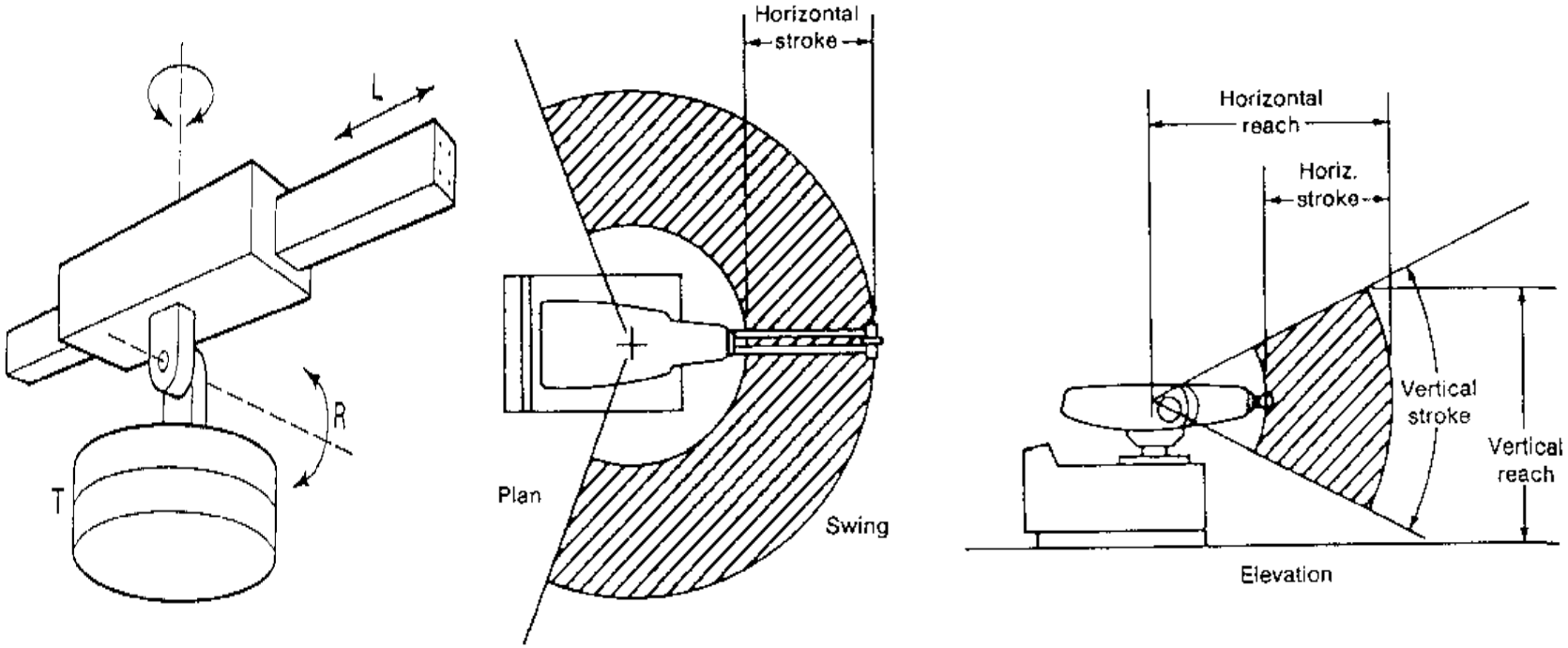


# Robot Configurations

## Spherical Robot - Work Space/Envelope

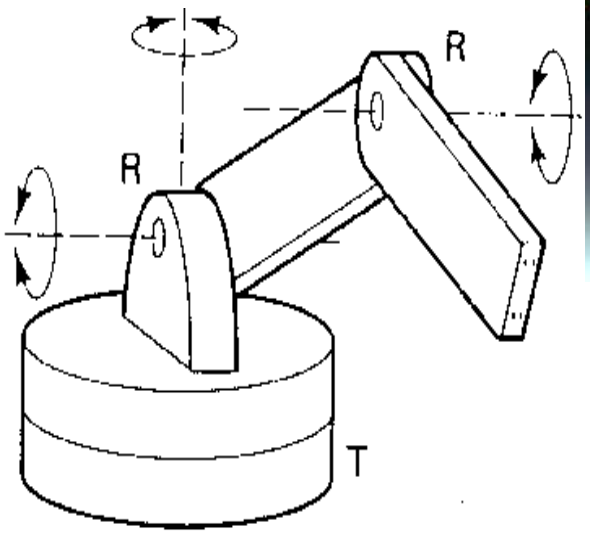
Spherical or polar robots are similar to a cylindrical robot, but form a spherical range of motion using a polar coordinate system.



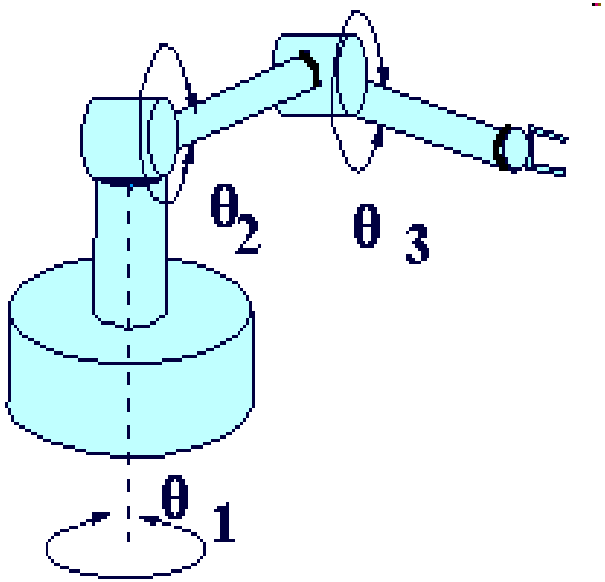
# Robot Configurations

**Articulated/anthropomorphic (3R/RRR) :**

An articulated robot's joints are all revolute, similar to a human's arm.





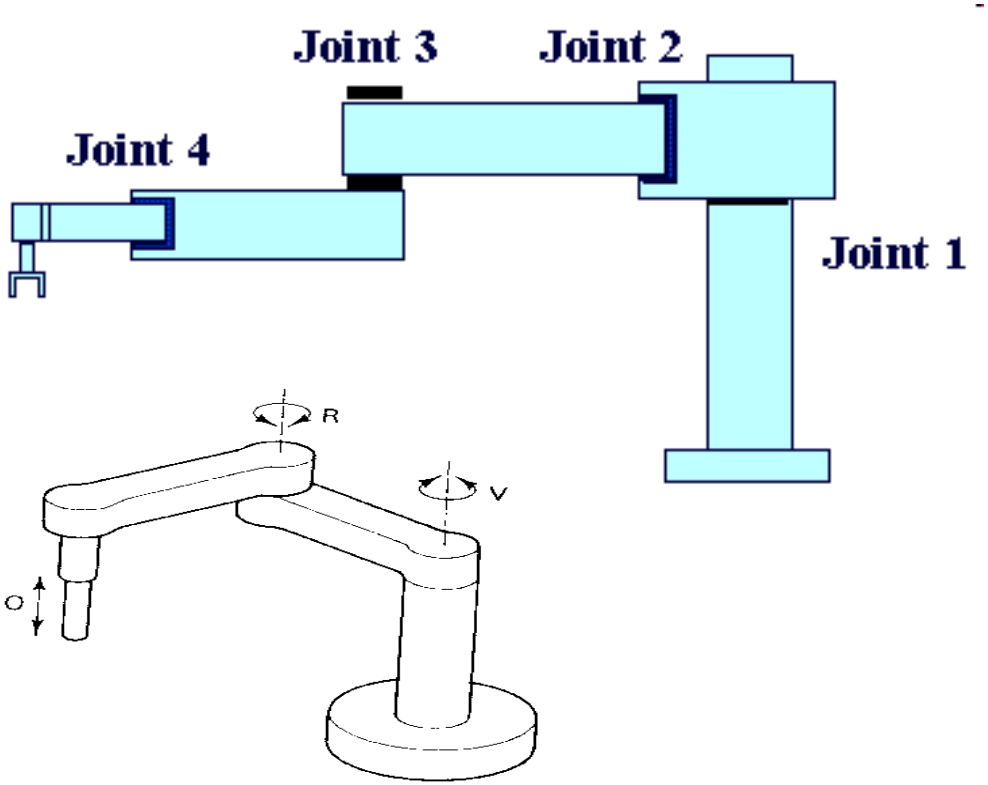



# Robot Configurations

**Selective Compliance Assembly Robot Arm (SCARA) (2R1P/RRP):**

They have two revolute joints that are parallel and allow the Robot to move in a horizontal plane, plus an additional prismatic joint that moves vertically



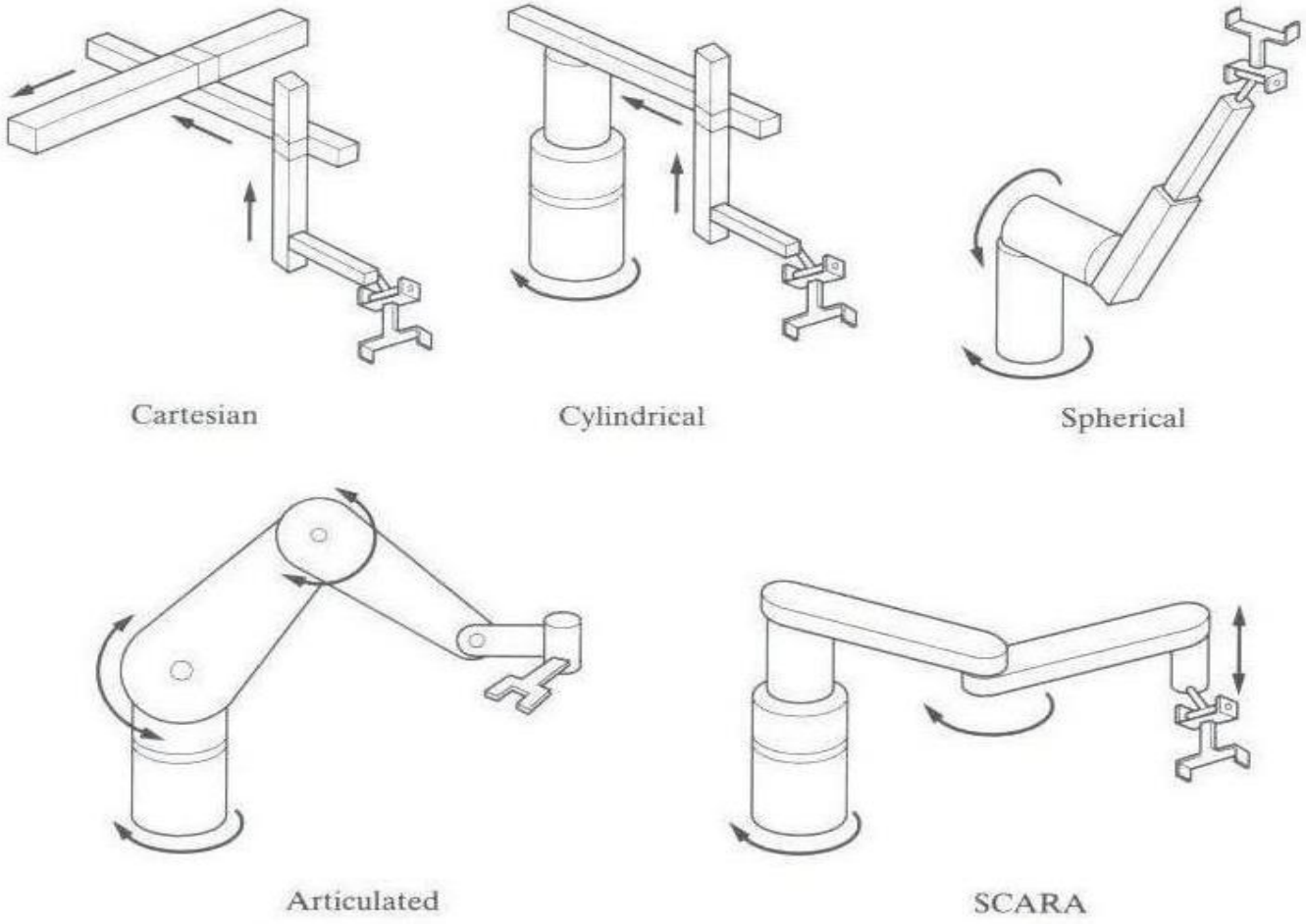
# Robot Configurations

**Parallel Robots**

Parallel Robots consist of a fixed base to a platform by means of a number of legs. This type of robot is used to create realistic flight simulators or rides in amusement parks.



# Robot Configurations



# Thank you for your attention !

## Good Luck !

